温州肯恩大学
WENZHOU-KEAN UNIVERSITY

2026SPW MGS 3001 WHS01
Python Programming for Business

Introduction to Regression

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Week 13-2, 21 May 2026

Where We Are

Date	Topic	Note
5/07	Structured Data Cleaning	✓ Pandas, duplicates, missing values, type conversion
5/12	Text Processing	✓ String method, regex, LLM-as-annotator
5/14	Exploratory Data Analysis	✓ Descriptive statistics, correlations, numeric vs. categorical
5/19	Data Visualization	✓ Matplotlib, Seaborn, visualization principles & practices
5/21	**Intro to Regression**	**TODAY – what regression does, how to interpret results**
5/26	Application of Regression	Run statistical models in Python, diagnostics, three-line table
5/28	Research Workshop III	Paper anatomy, section-by-section guidance, academic writing with AI
6/02	Research Workshop IV	Draft feedback, presentation preparation, consultation
6/04 & 09	Final Presentations	8-min presentation + 2-min Q&A per project

Final report due on June 10 (Wednesday) at 23:59

Where We Are

Research Question ▶ Hypothesis ▶ Data Collection ▶ Data Cleaning ▶ EDA & Visualization ▶ **Regression Analysis** ▶ Paper & Presentation

You've collected your data, cleaned it, explored patterns, and created visualizations.

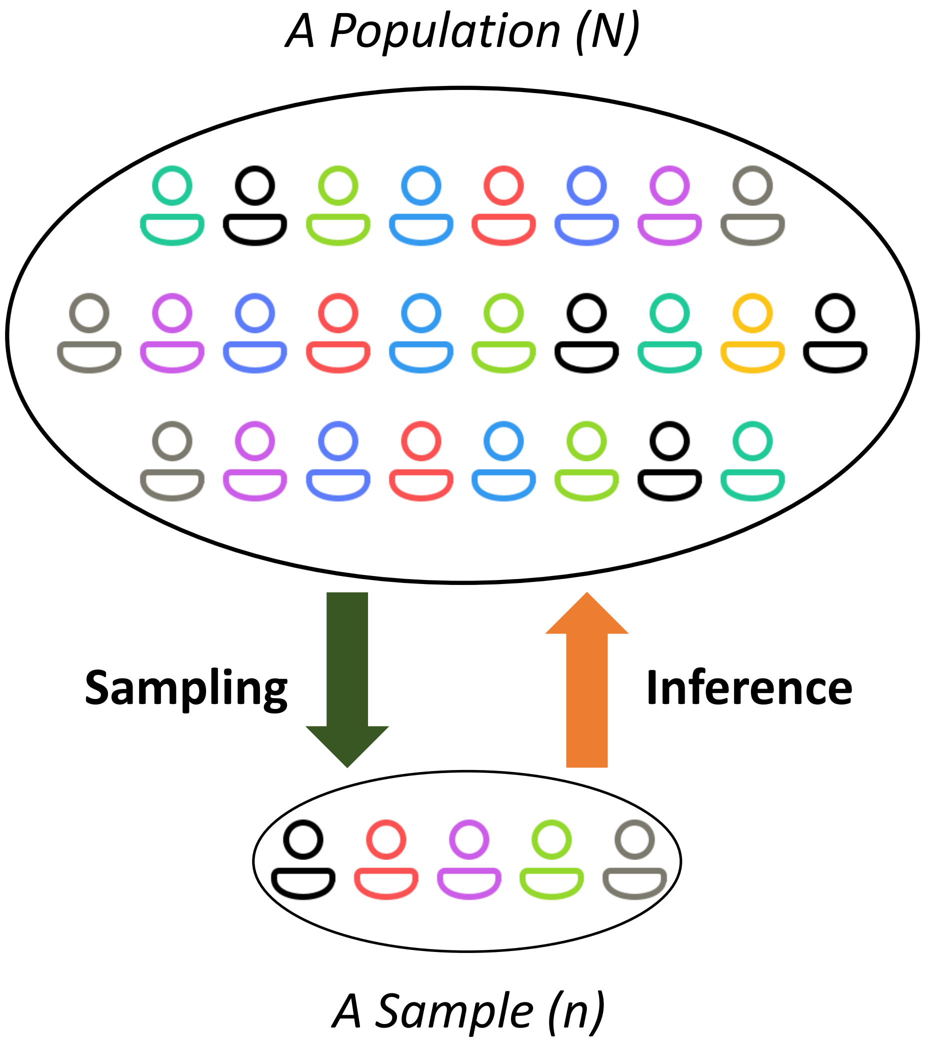
Today, we move from describing patterns to testing whether they are real.

EDA & Visualization	Statistical Inference	Paper & Presentation
What patterns exist?	*Is it likely true in general?*	*What do the findings mean?*

Statistical Inference

The Problem – Why Not Just Compare Averages?

- You surveyed 100 students — not all WKU students.
- Your sample shows **a positive relationship between study hours and GPA** (e.g., via scatter plot, correlation coefficient)
- But is this pattern real, or could it just be sampling noise? Would another 100 students show the same result?
- **Statistical inference** answer this: it tells you whether a pattern in your sample likely holds in the broader population.



Statistical Modeling

Input	Model	Output
Two groups (e.g., org vs. user repos)	**t-test, ANOVA**	Group difference (significant or not)
Numeric X and Y (e.g., study hours → GPA)	**Regression (OLS)**	Relationship direction, size, and significance
Numeric X, categorical Y (e.g., predict success/fail)	**Logistic regression, SVM, Random Forest, XGBoost**	Classification, prediction accuracy
Images, text, sequences, graphs	**Deep Learning (NN, CNN, Transformer, GNN)**	Complex pattern recognition, generation

- Different models for different data types, tasks, and assumptions.
- All serve the same purpose: find patterns in data and assess whether they are real.
- This is a Python course — we use regression to demonstrate how to run and interpret statistical models in Python.

Learning Objectives

By the end of this session, you should be able to:

1. Explain what regression analysis does and its logics
2. Interpret regression coefficients, p-values, and R^2 in plain language
3. Recognize how regression connects your hypothesis to statistical evidence
4. Understand what a regression table in a research paper communicates

Motivating Example

Do more study hours lead to higher GPA?

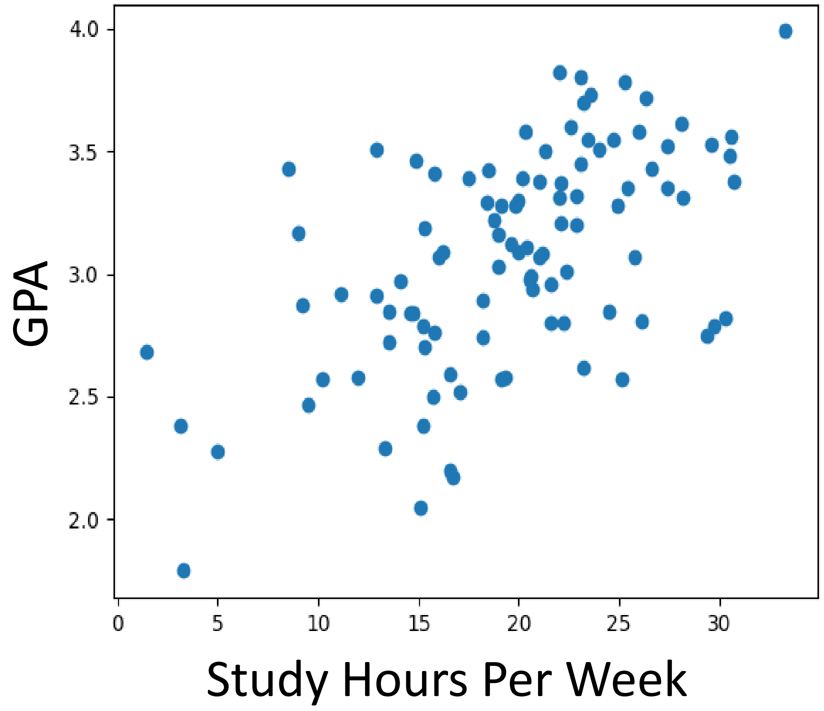
- **Scenario:** We want to study whether more study hours lead to higher GPA at WKU. We randomly selected 100 students and collected their weekly study hours and GPA.
- Each observation represents one student:
    - **Study Hours (X):** Average study hours per week
    - **GPA (Y):** overall grade point average (0 – 4.0 scale)

Study Hours	GPA
16.0	3.07
9.0	3.17
22.0	3.31
24.9	3.28
21.0	3.07

	Study Hours	GPA
mean	19.65	3.07
std	6.42	0.44
min	1.40	1.79
50%	20.25	3.09
max	33.30	3.99

Motivating Example

Do more study hours lead to higher GPA?



BONUS TIME!

What insights can you draw from this scatter plot?

- Can we predict a student's GPA based on their study time?
- On average, how much does GPA increase for each additional hour of study per week?
- How do we explain why a hard-working student may still not achieve a high GPA?

What Is Regression Analysis?

A statistical method for estimating the relationship between a dependent variable (DV) and one or more independent variables (IV) using observed data

Source: Wikipedia

What Regression Does

- ✓ Estimates the **direction** and **size** of a relationship
- ✓ Tests whether the relationship is statistically significant
- ✓ Controls for other variables simultaneously
- ✓ Produces results you can report in a paper

What Regression Does NOT Do

- ✗ Prove **causation** by itself
- ✗ Replace domain knowledge or theory
- ✗ Work well with garbage data
- ✗ Tell you which variables to include

Regression in Your Research Pipeline

Statistical modeling (e.g., regression) bridges your hypothesis (theory) to your evidence (empirical results), and ultimately to your implications and conclusions.

1 Hypothesis

Students who study more hours per week tend to have higher GPAs

2 Regression Model

GPA = β_0+ β_1 × StudyHours + β_2 × IQ + β_3 × SleepHours + ε

3 Evidence

Each additional hour of study per week is associated with a 0.05 increase in GPA on average.

- Your hypothesis tells you what to test.
- Regression tells you how to test it.
- The results tell your reader what you found.

The Regression Equation

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k + \varepsilon$$

$\boldsymbol{Y}$

Dependent Variable

What you want to explain or predict (e.g., GPA)

X

Independent Variable(s)

What you believe influences Y (e.g., study hour, IQ)

Control Variables

Other factors that may affect Y but are not the focus of your study. Included to isolate the effect of X on Y.

$\boldsymbol{\beta}_0$

Intercept

The predicted value of Y when all X = 0

$\boldsymbol{\beta}_1, \boldsymbol{\beta}_2, \ldots$

Coefficients

How much Y changes for a one-unit change in X

$\boldsymbol{\varepsilon}$

Error Term

What the model cannot explain

Simple vs. Multiple Regression

Simple Regression	Multiple Regression
$Y = \beta_0 + \beta_1 X + \varepsilon$	$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \varepsilon$
• One independent variable	• Multiple independent variables
• Good for exploring a single relationship	• Controls for confounders
• Quick to run and interpret	• Isolates each variable’s unique effect
• Does not control for other factors	• Required for most academic papers
E.g. GPA = $\beta_0 + \beta_1 \times$ StudyHours + ε	E.g. GPA = $\beta_0 + \beta_1 \times$ StudyHours + $\beta_2 \times$ IQ + $\beta_3 \times$ SleepHours + ε

The Regression Equation

Let's practice with a different example.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k + \varepsilon$$

	House price	Size	City	…	Year
	$\boldsymbol{Y}$	$\boldsymbol{X_1}$	$\boldsymbol{X_2}$	**…**	$\boldsymbol{X_k}$
House 1	y_1	x_{11}	x_{21}	…	x_{k1}
House 2	y_2	x_{12}	x_{22}	…	x_{k2}
	…	…	…	…	…
House n	y_n	x_{1n}	x_{2n}	…	x_{kn}
House m	?	x_{1m}	x_{2m}	…	x_{km}

You want to predict house price.

- What is the dependent variable? Also called outcome, response, or DV
- What are the independent variables? Also called predictor, feature, or IV
- What would be good control variables?

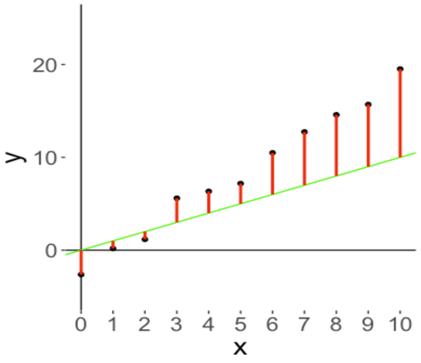
Coefficients Estimation

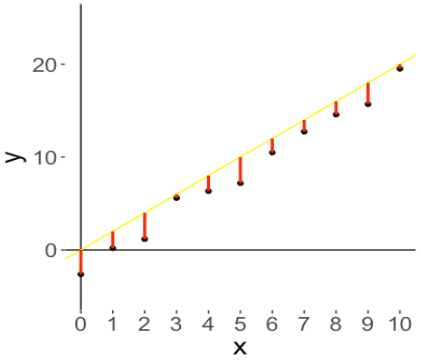
- You have data on Y and $X_1, X_2, \ldots, X_K$ for each observation in your sample.
- The parameters / coefficients $(\beta_0, \beta_1, \beta_2, \ldots, \beta_k)$ are unknown.
- In other words, the pattern (represent by the regression equation) is unknown.
- **You goal:** learn the pattern (i.e., estimate the parameters / coefficients) from data. How?

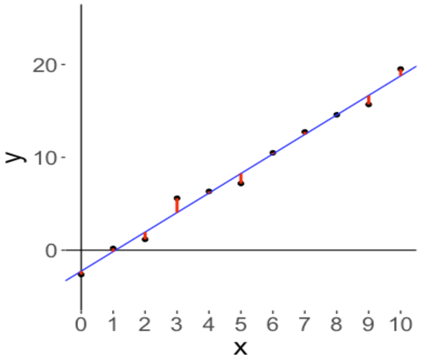
"True" model: $y = \beta_0 + \beta_1 x + \varepsilon$

Our model: $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$

Residual / Residual Error: $e_i = y_i - \hat{y}_i$







Coefficients Estimation – Ordinary Least Squares (OLS)

Goal: Choose $\hat{\beta}_0, \hat{\beta}_1$ so as to **minimize the sum of squares of the residuals**.

$$\sum_i^n e_i^2 = \sum_i^n (y_i - \hat{y}_i)^2 = \sum_i^n (y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i))^2$$

Derivation (for your edification, not tested): Minimize by setting derivatives to 0.

Define means: $\bar{y} \equiv \frac{1}{n}\sum_i^n y_i,\ \bar{x} \equiv \frac{1}{n}\sum_i^n x_i$

$$Z \equiv \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

$$\frac{\partial Z}{\partial \hat{\beta}_0} = \sum_{i=1}^n -2(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

$$\frac{\partial Z}{\partial \hat{\beta}_1} = \sum_{i=1}^n -2x_i(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

$$\hat{\beta}_0 = \frac{1}{n}\sum_{i=1}^n y_i - \frac{1}{n}\sum_{i=1}^n \hat{\beta}_1 x_i \Rightarrow \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

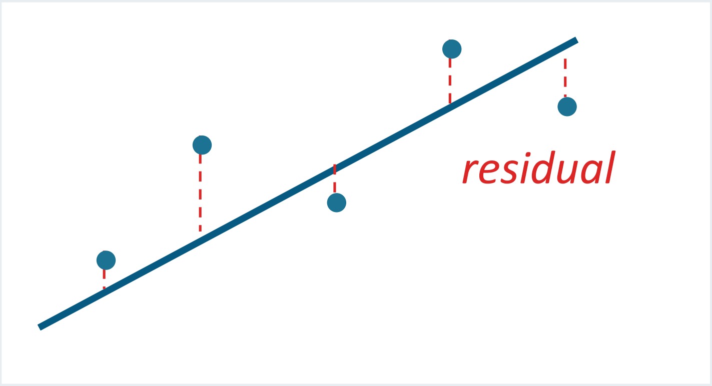
$$\sum_{i=1}^n x_i y_i - \sum_{i=1}^n (\bar{y} - \hat{\beta}_1 \bar{x}) x_i - \sum_{i=1}^n \hat{\beta}_1 x_i^2 = 0$$

$$\sum_{i=1}^n x_i y_i - \bar{y}\sum_{i=1}^n x_i + \hat{\beta}_1\left(\bar{x}\sum_{i=1}^n x_i - \sum_{i=1}^n x_i^2\right) = 0$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n x_i y_i - \bar{y}\sum_{i=1}^n x_i}{\sum_{i=1}^n x_i^2 - \bar{x}\ \sum_{i=1}^n x_i} = \frac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{\sum_{i=1}^n x_i^2 - n\bar{x}^2} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Coefficients Estimation – Ordinary Least Squares (OLS)

OLS finds the line that best fits the data by minimizing the total squared prediction errors.



Intuition

- The red dashed lines are residuals — the prediction errors.
- OLS squares each residual and adds them up.
- It picks the line that makes that sum as small as possible.
- You do not compute this yourself — Python does it for you. Your job is to set up the model correctly and interpret the output.

Interpreting Coefficients

$$\widehat{GPA} = \hat{\beta}_0 + \hat{\beta}_1 * \text{StudyHours} + \hat{\beta}_2 * \text{IQ} + \hat{\beta}_3 * \text{SleepHours}$$

The coefficient tells you the estimated effect size and direction.

> *"A one-unit increase in [X] is associated with [β] change in [Y] on average, holding other variables constant."*

IVs	**β**	**Interpretation**
Study Hours	**0.05**	Each additional hour of study per week is associated with 0.05 increase in GPA **on average**
IQ	**0.02**	Each additional IQ point is associated with 0.02 increase in GPA **on average**
Sleep Hours	**0.08**	Each additional hours of sleep per night is associated with a 0.08 increase in GPA **on average**

⚠ "Associated with" ≠ "caused by." Regression shows correlation with controls, not causation.

Statistical Significance: p-Values

Core question:

Is the relationship I found likely to be real, or could it have appeared by chance?

What the p-Value Tells You

The probability of seeing a coefficient this large (or larger) if the true effect were zero.

● **$p < 0.001$**	***	Very strong evidence
● **$p < 0.01$**	**	Strong evidence
● **$p < 0.05$**	*	Moderate evidence
● **$p \geq 0.05$**		Not significant

Common Misinterpretations

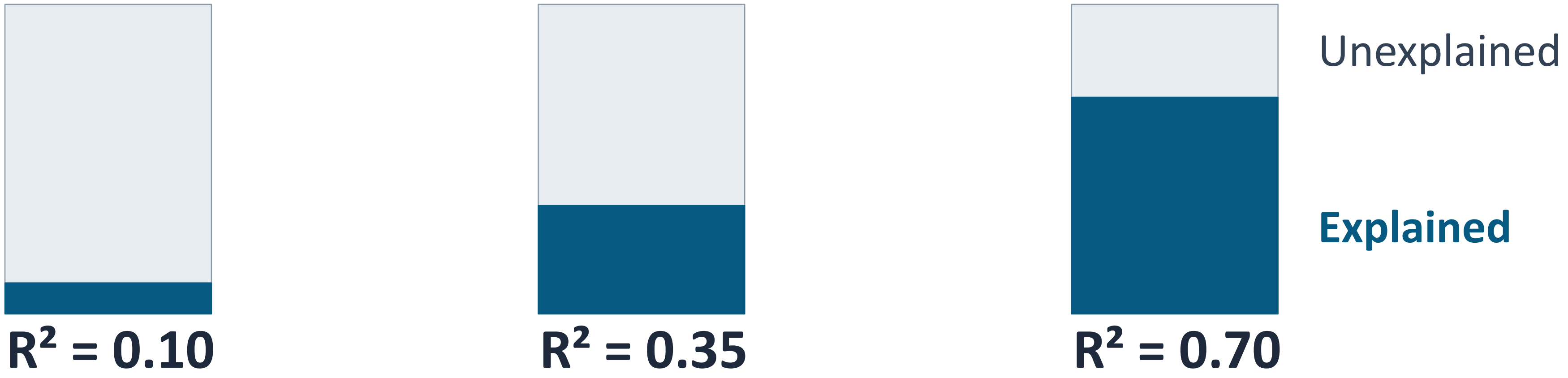
✗ "$p < 0.05$ means the result is true."
✓ It means the result is unlikely under the null hypothesis.

✗ "$p = 0.06$ means no effect."
✓ It means the evidence is weaker, not absent.

✗ "Smaller p = bigger effect."
✓ p-value reflects certainty, not size. Report β for effect size.

Model Fit: R^2 (R-Squared)

R^2 = the proportion of variance in Y that your model explains. It ranges from 0 to 1.



- A higher R^2 is generally desirable, but there is no universal threshold for what counts as acceptable.
- Avoid inflating R^2 by adding irrelevant variables.

Reading a Regression Table

Table 1. Regression Results

	Model 1	**Model 2**	**Model 3**
$StudyHours$	0.052^{***}	0.048^{***}	0.043^{***}
	(0.012)	(0.011)	(0.013)
IQ		0.019^{***}	0.017^{**}
		(0.006)	(0.006)
$SleepHours$			0.081^{*}
			(0.035)
Constant	1.820^{***}	0.950^{**}	0.420
	(0.210)	(0.340)	(0.380)
R^2	0.180	0.250	0.291
N	200	200	200

** p < 0.05, ** p < 0.01, *** p < 0.001*

Note: Standard errors in parentheses. Dependent variable is GPA.

Regression results are typically reported using three-line tables, though formats vary across journals and disciplines.

Shan and Qiu (2025)

Demirci et al. (2025)

Why Control Variables Matter

The Omitted Variable Problem

Without controls: study hours → GPA (β = 0.052***)	*Looks strong, but is study hours the whole story?*
With controls: study hours → GPA (β = 0.043***)	*Effect shrinks — part was due to IQ and sleep.*

What happened?

- Students who study more may also sleep better or have other habits that boosts GPA.
- Without controlling for these factors, the study hours coefficient absorbs effects that belong to other variables.
- Controls isolate each variable's unique contribution, making your estimates more credible.

Model Specification: What to Include

Choosing which variables to include is a research decision, not a technical one.

Include If...

- Theory or prior research suggests it affects Y
- It could confound the X→Y relationship
- It is available in your dataset and measured reliably

Exclude If...

- No theoretical justification
- It is a consequence of Y, not a cause (mediator/outcome)
- Including it would cause multicollinearity

Be Careful If...

- Two predictors are highly correlated (> 0.8)
- You have more variables than observations
- Adding a variable changes other coefficients dramatically

Advanced topics — we won't cover

Quadratic term?
Interaction term?

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 (X_1 \times X_2) + \beta_4 X_1^2 + \varepsilon$$

Next Session: Applied Regression with Python

May 26 — Hands-on regression analysis using your project data

- Load & inspect your cleaned data
- Specify and run OLS regression using statsmodels
- Interpret the output table
- Run diagnostics and check assumptions
- Export publication-ready regression tables

Before next session: Make sure your cleaned dataset is ready. Review your hypothesis and identify your DV, main IV, and control variables.

Q&A

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Office hours: Tue & Thu 11:15–12:30 | Wed 14:00–16:30